

# Hypothesis 3 – Relationship Between Daily Steps and Sleep Duration

## 1. Introduction

This analysis examines whether there is a significant association between **Daily Steps** and **Sleep Duration** among participants. Understanding this relationship helps determine whether physical activity is linked to sleep behavior.

### Hypotheses

- **Null Hypothesis ( $H_0$ ):** There is no correlation between daily steps and sleep duration.
  - **Alternative Hypothesis ( $H_4$ ):** There is a significant correlation between daily steps and sleep duration.
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## 2. Descriptive Statistics

Descriptive statistics for the variables are as follows:

| Variable             | Count | Mean    | Std     | Min  | Max   |
|----------------------|-------|---------|---------|------|-------|
| Daily Steps          | 374   | 6816.84 | 1617.92 | 3000 | 10000 |
| Sleep Duration (hrs) | 374   | 7.13    | 0.80    | 5.8  | 8.5   |

These statistics show that the dataset is reasonably spread out for steps and has relatively stable sleep duration.

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## 3. Normality Check

A Shapiro–Wilk test was conducted to assess normality of both variables:

- **Daily Steps:** p-value =  $1.65 \times 10^{-11}$
- **Sleep Duration:** p-value =  $1.27 \times 10^{-11}$

Both variables deviate significantly from normality ( $p < 0.05$ ). Therefore, **Spearman correlation**, a non-parametric test, was used.

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## 4. Correlation Analysis

The Spearman rank-order correlation was calculated:

- **Spearman correlation ( $\rho$ ):** 0.01
- **p-value:** 0.846

**Interpretation:**

- $\rho \approx 0 \rightarrow$  extremely weak correlation
- p-value > 0.05  $\rightarrow$  not statistically significant

This indicates **no meaningful monotonic relationship** between daily steps and sleep duration.

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## 5. Visualization

To support the analysis, the following plots were generated:

1. **Scatter Plot with Regression Line:** Shows a random scatter of points and a nearly flat trend line, indicating no visible relationship.
2. **Spearman Correlation Heatmap:** Clearly shows the correlation value near 0.

These visualizations corroborate the numerical results.

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## 6. Conclusion

Based on descriptive statistics, normality testing, Spearman correlation, and visualization:

- There is **no evidence of a significant relationship** between daily steps and sleep duration.
- The null hypothesis is **retained**.  $H_0$  is Accepted.

**Final Statement:**

The number of daily steps does **not** significantly predict sleep duration among participants in this dataset. Therefore, increasing or decreasing steps alone does not appear to affect the number of hours slept.